Environmental Protection Act 1986

Section 39A(7)

PUBLIC ADVICE

Proposal: Albany Artificial Surf Reef

Proponent: City of Albany

Decision: **Referral Examined, preliminary investigations and inquiries conducted. Proposal not to be assessed under Part IV of the Environmental Protection Act 1986 (EP Act)- Advice Given.**

The EPA considers that the proposal is unlikely to have a significant impact on the environment and does not warrant formal assessment.

Background:
On 28 September 2018, the City of Albany (CoA) referred a proposal to construct an artificial surfing reef (ASR) approximately 300 meters (m) offshore at Middleton Beach, southern Western Australia. The proposed ASR will be built at the southern end of the beach with the aim of enhancing the recreational surfing conditions for local residents and tourists.

It is proposed to construct the ASR using barge-mounted equipment instead of shore based construction methods. Barges, towed using local tugboats, will be used to transport rock material from nearby Albany Port. Construction material is likely to be locally sourced rock, approximately 800-1200mm in diameter. Approximately 50,000 m^3 of rock material will be transported to Albany Port by road haulage to construct the proposed ASR.

The proposal was advertised for public comment, and the EPA notes that 8 public comments were received. Key issues raised in the public comments included: impacts to marine fauna, impacts to sheltered and calm water, impacts to coastal processes and ongoing maintenance costs, financial cost, poor wave quality of previous artificial reefs (Cables Station Cottesloe), concerns regarding the stability of the seabed, improvement of the marine environment and surf quality at Middleton beach.

At the time of referral, the EPA considered that it did not have enough information regarding the potential impacts of the proposal to make a decision as to whether or not to formally assess, and if so the level of assessment. The EPA requested further information from the proponent in response to issues raised in technical advice

received from the Department of Transport (DoT). Information was received by the EPA on 4 February 2019 and 23 May 2019.

Relevant Statutory and Administrative Provisions
The EPA has considered the proposal in accordance with the requirements of the EP Act and the *Environmental Impact Assessment (Part IV Divisions 1 and 2) Administrative Procedures 2016* and *Procedures Manual*.

Materials considered in making this decision
The EPA has considered and had regard to the information:

- provided in the proponent's referral information, which is available on the EPA's consultation hub;
- derived from comments received during the 7-day comment period (8 comments were received);
- set out in the advice from the Department of Transport (DoT);
- derived from the EPA's own inquiries; and
- in the proponent's response to the EPA's requests for further information.

Consideration
In making its decision on whether to assess the proposal, the EPA had regard to various matters, including the following (as outlined in the EPA's *Statement of Environmental Principles, Factors and Objectives*):

a) values, sensitivity and quality of the environment which is likely to be impacted
b) extent (intensity, duration, magnitude and geographic footprint) of the likely impacts
c) consequence of the likely impacts (or change)
d) resilience of the environment to cope with the impacts or change
e) cumulative impact with other existing or reasonably foreseeable activities, developments and land uses
f) connections and interactions between parts of the environment to inform a holistic view of impacts to the whole environment
g) level of confidence in the prediction of impacts and the success of proposed mitigation
h) public interest about the likely effect of the proposal or scheme, if implemented, on the environment and public information that informs the EPA's assessment

In considering the potential direct and indirect impacts of the proposal on the **Benthic Communities and Habitat**, **Coastal Processes, Marine Environmental Quality** and **Marine Fauna**, the EPA has had particular regard to:

- the small scale of the proposal;
- the environmental values within and surrounding the proposal area;
- the low level of public interest in this proposal and issues relating to the environmental factors;
- the community and stakeholder consultation undertaken to date by the proponent;
- the small scale of predicted impacts to Benthic Communities and Habitat at Middleton beach;
- the adjoining Oyster Harbour in the north and Princess Royal Harbour to the south both contain extensive ecologically important seagrass communities;
- seagrass meadows in exposed environments like much of King George Sound are highly dynamic, with patches moving over 10 to 50 year time periods
- historical storm events in the area scouring the seabed and dislodging seagrass, decreasing seagrass meadows along Middleton beach;
- the proponents adoption of a worst case scenario approach to impact prediction, which encompasses an area of seabed that would be occupied if all Artificial Surf Reef (ASR) design options currently considered were built, representing a highly conservative method of predicting potential impacts of the ASR on Benthic Communities and Habitat;
- the current preferred design option is sited entirely within an area mapped as wrack;
- the predicted low impact to coastal processes at Middleton Beach from modelling undertaken to date;
- the proponents commitment to undertake further detailed design and modelling to determine final design parameters and further understand potential impacts to coastal processes prior to construction commencing.
- the material for the ASR structure will be inert;
- construction activities have potential to cause turbidity generation, spills and waste, but are short and temporary in duration, including no piling requirements;
- the use of slow moving construction vessels reducing risk of fauna strike;
- historical sampling indicates that King George Sound sediments and waters are generally of a high quality;
- there are no recorded nesting sites for turtles in southern Western Australia as sand temperatures are inadequate for successful incubation;
- there are no significant cumulative impacts when considering the surrounding use of the coastal environment;
- the mitigation strategies proposed by the proponent to avoid and minimise impacts, which include:
    - implementation of a proposal specific construction Environmental Management Plan during construction activities including management

measures to minimise loss of Benthic Communities and Habitat, daily plume sketches and site photos, daily inspections of hydrocarbon equipment, spill and rubbish inspections, construction equipment and related vessels to be cleaned prior to mobilisation, all vessels regulatory introduced marine species procedures and requirements and a dedicated marine fauna observer on board construction vessels and or equipment; and

- implementation of a proposal specific Operational Environmental Management Plan (currently in draft state with final design options still being considered) during operations outlining coastal process monitoring, management and contingency measures during operations of the proposal.

In addition to the above, the EPA notes the presence of other statutory processes that can manage the potential impacts, including:

- management of maritime activities by the DoT in Western Australian navigable waters under the *Navigable Waters Regulations 1958* (referring to the *Shipping and Pilotage Act 1967*, *Jetties Act 1926* and *Western Australian Marine Act 1982*);
- a construction permit from Department of Water and Environmental Regulation under the *Waterways Conservation Act 1976* and *Waterways Conservation Regulations 1981*; and
- regulations for ballast water; biofouling and biosecurity risks associated with marine pests under the Commonwealth *Biosecurity Act 2015* administered by the Department of Agriculture and Water Resources.

In summary, although the proposal raises a number of environmental issues, the EPA considers that its objectives for **Benthic Communities and Habitat, Coastal Processes, Marine Environmental Quality and Marine Fauna** can be met.

The EPA is of the view that the potential impacts can be adequately managed through the implementation of the proposal in accordance with the proponents referral documentation (which includes the proponent's monitoring, management and mitigation measures), implementation of the EPA's advice given to other relevant decision making authorities and dealt with by other statutory processes. As a result, the EPA considers that the likely environmental effects of the proposal are not so significant as to warrant formal assessment.

1. Advice and Recommendations regarding Environmental Issues

The EPA notes the additional work done by the proponent, and the advice from the DoT, to prepare the *Amazing South Coast Surf Reef Draft Operational Environmental Management Plan* (22 May 2019) (prepared by Bluecoast Consulting Engineers Pty Ltd) (the Draft OEMP).

The EPA notes the Draft OEMP is focused on coastal processes and outlines monitoring, management and contingency measures during operations of the proposal. The development of the Draft OEMP was requested to provide more

confidence that the potential environmental impacts to coastal processes can be managed to an acceptable level over the short, medium, and long term during the proposals operational phase.

The EPA notes the artificial reef is currently in the design phase and that there are multiple design options being considered. The EPA notes the proponent has committed to undertake additional detailed design and modelling of the proposed ASR during the detailed design phase to determine final configuration and confirm predicted impacts to coastal processes will be localised, prior to construction commencing. The EPA considers the additional design and modelling work will confirm potential impacts are as currently predicted in the proponent's referral documentation.

Based on the above, the EPA recommends that the proponent:

- implement the action in the referral information about preparing detailed designs and further coastal process modelling once a configuration is chosen;
- continue to advance the draft OEMP to a Final OEMP based on the further detailed designs and coastal process modelling, and to further confirm the proposal can be adequately managed during the operational phase; and
- refer the final design(s), modelling and OEMP to the DoT for their review and endorsement prior to DWER considering an application under the *Waterways Conservation Act 1976* and prior to commencement of construction activities.